

PROJECT QUALITY MANAGEMENT

A magnifying glass with a silver frame and a purple lens is positioned on the left side of the slide. The lens is focused on the word 'QUALITY' in the title 'PROJECT QUALITY MANAGEMENT'. The handle of the magnifying glass extends downwards and to the right. A thin horizontal line is positioned below the title text.



Session objectives

- Explain quality in software project management
- Explain project quality management processes
- Describe tools and techniques
- Explain the types of project quality costs



What is Project Quality Management?

- The International Organization for Standardization (ISO) defines quality as the totality of characteristics of an **entity** that determine its ability to satisfy stated or implied needs.
- Other experts define quality based on
 - **Conformance to requirements**: meeting written specifications
 - **Fitness for use**: ensuring a product can be used as it was intended



Project Quality Management Processes

1. **Quality planning: identifying** which quality standards are relevant to the project and how to satisfy them
2. **Quality assurance: evaluating** overall project performance to ensure the project will satisfy the relevant quality standards
3. **Quality control: monitoring** specific project results to ensure that they comply with the relevant quality standards while identifying ways to improve overall quality



Modern Quality Management

- Modern quality management
 - requires customer satisfaction
 - prefers prevention to inspection
 - recognizes management responsibility



Quality Planning

- It is important to design in quality and communicate important factors that directly contribute to **meeting the customer's requirements**
- Design of experiments helps identify **which variables have the most influence** on the overall outcome of a process
- Many scope aspects of software projects affect quality, like functionality, features, system outputs, performance, reliability, and maintainability



Quality Assurance

- Quality assurance includes all the activities related to **satisfying the relevant quality standards** for a project
- Another goal of quality assurance is **continuous quality improvement**
- **Benchmarking** can be used to generate ideas for quality improvements
- **Quality audits** help identify lessons learned that can improve performance on current or future projects



Quality Control

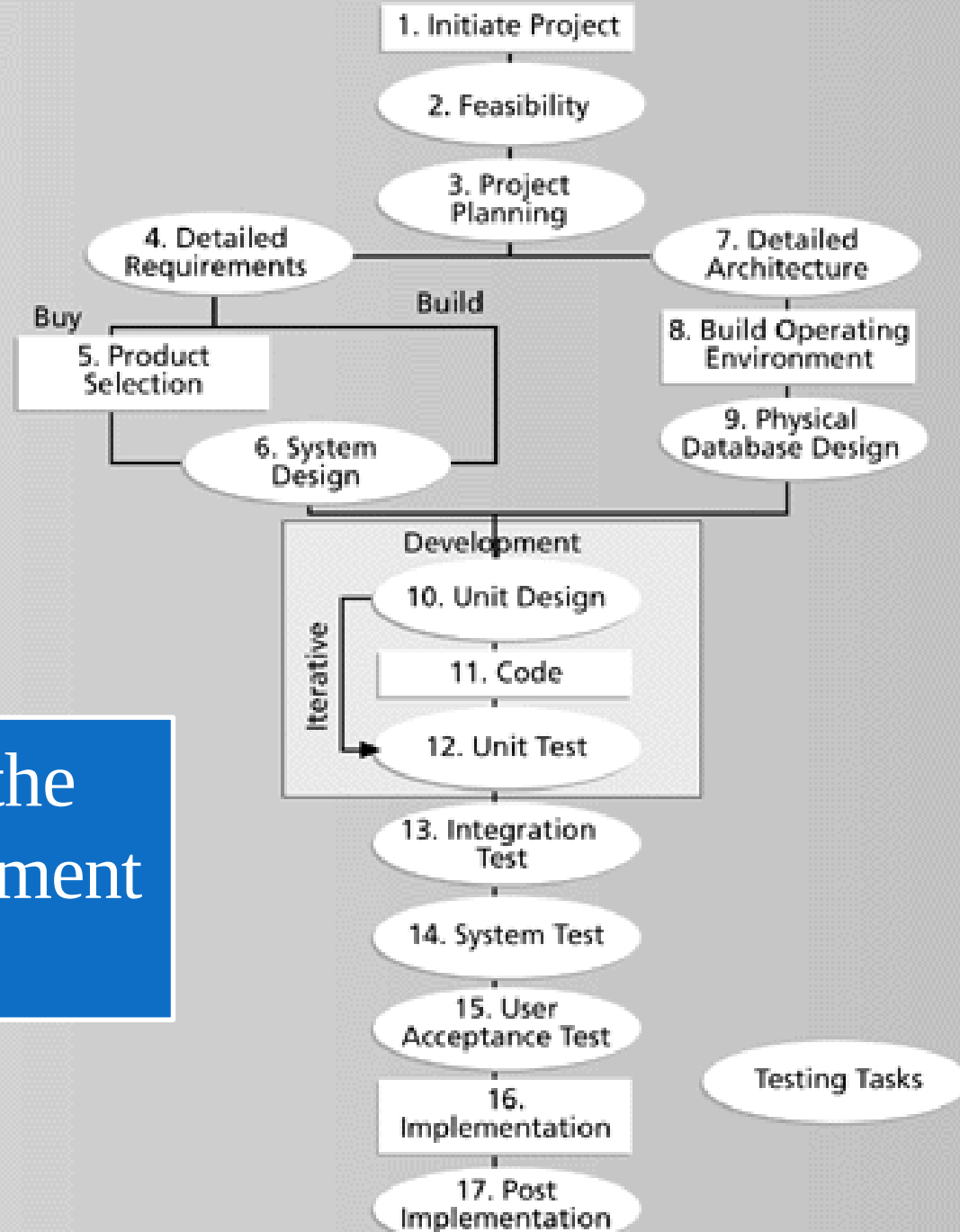
- The main outputs of quality control are
 - Acceptance decisions
 - Rework
 - Process adjustments
- Some tools and techniques include
 - Pareto analysis
 - Statistical sampling
 - Quality control charts
 - Testing



Testing

- Many SW professionals think of testing as a stage that comes near the end of SW product development
- Testing should be done in almost every place of the SW product development life cycle

Testing tasks in the software development life cycle

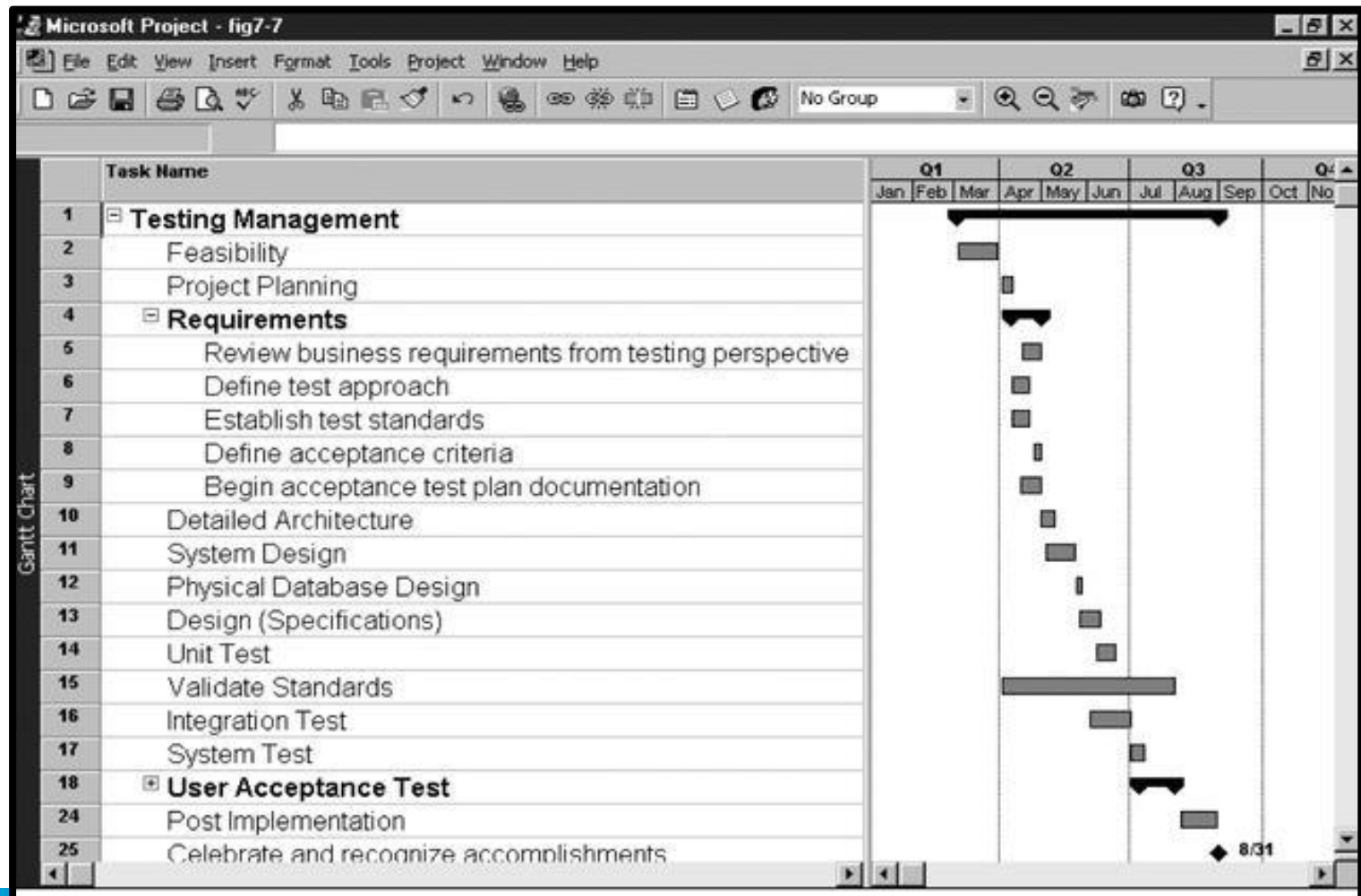




Types of Tests

- **Unit testing** is done to test each component to ensure it is as defect-free as possible
- **Integration testing** occurs between unit and system testing to test functionally grouped components
- **System testing** tests the entire system as one entity
- **User acceptance testing** is an independent test performed by the end user before accepting the delivered system

Gantt Chart for Building Testing into a Systems Development Project Plan





Improving Software Project Quality

- Several suggestions for improving quality for Software projects include:
 - Leadership that promotes quality
 - Understanding the cost of quality
 - Focusing on organizational influences and workplace factors that affect quality
 - Following maturity models to improve quality



Leadership

“It is most important that top management be quality-minded. A large percentage of quality problems are associated with management, not technical issues



The Cost of Quality

- The cost of quality is
 - the cost of **conformance** or delivering products ~~to~~ meet requirements and fitness for use
 - the cost of **nonconformance** or taking responsibility for failures or not meeting quality expectations



Cost Categories Related to Quality

➤ The Cost of Quality category codes are the following:

- 1) Prevention Costs
- 2) Appraisal Costs
- 3) Internal Error Costs
- 4) External Error Costs
- 5) Measurement and test equipment costs



1. Prevention Costs

- Prevention costs are investments ahead of time in an effort to ensure conformance to requirements.
- Examples include activities such as orientation of team members, training, and the development of project standards and procedures.



2. Appraisal Costs

- Appraisal costs are costs acquired to identify defects after the fact.

Examples include activities such as **walk-throughs** and **testing**.



3. Internal Error Costs

- Internal error costs are the costs of **rework** and **repair** before delivery to a customer.

An example is **fixing faults detected during internal testing.**



4. External Error Costs

- External error costs are the costs of ~~rework~~ and repair after delivery to a customer.
- One example would be **rework and repair** resulting from acceptance testing.
- Another example would be the actual ~~costs~~ incurred during **warranty support**.



5. Measurement and Test Equipment Costs

- Measurement and test equipment costs include the capital cost of equipment used to perform prevention and appraisal activities



ISO 9000

- ▮ An international set of standards for quality management.
- ▮ Applicable to a range of organisations from manufacturing to service industries.
- ▮ ISO 9001 applicable to organisations who design, develop and maintain products.
- ▮ ISO 9001 is a generic model of the quality process that must be instantiated for each organisation using the standard.



ISO 9000 certification

- Quality standards and procedures should be documented in an organisational quality manual.
- An external body may certify that an organisation's quality manual conforms to ISO 9000 standards.
- Some customers require suppliers to be ISO 9000 certified, although the need for flexibility here is increasingly recognised.



Software quality attributes

Safety	Understandability	Portability
Security	Testability	Usability
Reliability	Adaptability	Reusability
Resilience	Modularity	Efficiency
Robustness	Complexity	Learnability

Many thanks

